Khushal Agrawal

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Software Engineer - Data Infrastructure, Databases, Distributed Systems, Python/Rust/Linux

Education

Indian Institute of Technology Roorkee Oct 2022 – June 2026
B.Tech in Computer Science and Engineering GPA: 8.8/10.0
Relevant Coursework: Operating Systems, Computer Architecture, Compiler Design, Database Management Systems, Systems Software, Data Structures and Algorithms, Network Theory, Artificial Intelligence

Experience

Ambient Security Dec 2025 – Present
Software Engineer *Remote*

- Work on infrastructure for petabyte-scale sensitive security data, using Azure Event Hub and JanusGraph for event ingestion, graph modeling, and large-scale analysis workflows.
- Optimize graph algorithms over large security datasets to improve query and analysis speed while maintaining detection accuracy.
- Set up a Kong API gateway on an AKS cluster to support secure service routing and cloud infrastructure deployment.
- Built a SailPoint ingestion connector and wrote end-to-end integration tests for a large security data workflow.

Goldman Sachs June 2025 – Aug 2025
Summer Analyst *Bengaluru, India*

- Designed and implemented a data migration tool for petabyte-scale in-house datasets to Snowflake using Apache Flink and Hadoop-backed processing.
- Built Java/Jersey microservices and React status views to track migration progress, surface operational state, and improve visibility for internal data engineering users.
- Developed BPMN workflows to simplify migration operations and reduce manual coordination across data platform users.

Uber AI Jan 2026 – Present
AI Prompt Engineer for Google's Gemini *Remote*

- Designed evaluation tasks for Gemini on Terminal Bench and Colab Bench, targeting temporal reasoning, multimodal data integration, anomaly analysis, and physics-based inference.
- Built Python, Jupyter, Docker, NumPy, and SciPy test harnesses to validate multi-step agent trajectories and intermediate artifacts.

Google Summer of Code, Inkscape May 2024 – Aug 2024
Open Source Developer *Remote*

- Reworked the Node and Pen Tool workflow in a large C++/Gtkmm open-source codebase, reducing tool-switching friction for vector graphics editing.
- Optimized Bezier curve rendering paths through algorithmic improvements and performance-oriented C++ changes.

Lysto.io Feb 2025 – Jan 2026
Software Engineer *Remote*

- Automated daily market analysis workflows using n8n, Firecrawl, Supabase, Tailwind, and Next.js for data extraction, storage, and user-facing research flows.

Projects

Redis Clone github.com/SK1PPR/redis
Rust, Tokio, TCP, RESP, async networking, performance benchmarking

- Built a Redis-compatible in-memory key-value server with RESP parsing, async TCP networking, string commands, pub/sub, streams, sorted sets, and geospatial commands.
- Benchmarked supported workloads against Redis 8.6.2 and reached approximately 95% of Redis performance after profiling and removing pathological scans in hot paths.
- Documented benchmark methodology, profiling results, side-index optimizations, latency/throughput tradeoffs, and remaining gaps versus production Redis.

Kafka Clone github.com/SK1PPR/kafka
C++, distributed log, broker APIs, persistence, tests

- Built a Kafka-like broker with CreateTopics, Produce, Fetch, and ListOffsets APIs, append-only topic-partition logs, byte-offset reads, and filesystem-backed persistence.

- Implemented offset progression, earliest/latest offset lookup, fetch-from-offset behavior, duplicate topic handling, and broker API tests for message flow correctness.
- Designed the project around distributed log fundamentals, producer/consumer semantics, partition metadata, and scalable streaming infrastructure.

VortexDB github.com/sdslabs/VortexDB

Rust, RocksDB, in-memory storage, vector search, ANN benchmarking

- Built a Rust vector database engine with pluggable storage backends, including RocksDB-backed persistence and an in-memory storage engine.
- Implemented embedding generation hooks, KD-tree indexing, and KNN search for multimodal vector retrieval workloads.
- Preparing the database for ANN benchmark submission, focusing on recall, query latency, index behavior, memory overhead, and storage tradeoffs.

C Compiler for MIPS github.com/sanatan01/CSN-352-Project

C/C++, compiler design, parsing, code generation, MIPS

- Built a C compiler targeting the MIPS instruction set, implementing lexical analysis, parsing, semantic checks, code generation, and assembly emission.
- Implemented lowering for expressions, control flow, function calls, stack frames, and generated MIPS programs.

Achievements

- Gold Medal, Inter-IIT Tech Meet 2023, IGDC problem statement; Bronze Medal as PS-Lead, Inter-IIT Tech Meet 2024, IGDC problem statement.
- 2nd in India, CSAW Embedded Security Challenge; Winner, Alphagrep Scholarship AY 2024.
- Codeforces rating: 1592; JEE Advanced 2022 rank: 337 among 150,000 candidates.

Technologies

Languages: Python, Rust, C++, C, Java, SQL, Bash, JavaScript/TypeScript
Data and Systems: PostgreSQL, Kafka, Redis, RocksDB, Flink, Hadoop, Snowflake, JanusGraph, Azure Event Hub, data mining, ETL, web crawling, graph algorithms, storage engines, stream processing
Tools: Linux, Git, Docker, GDB, perf, Nix basics, Jupyter, NumPy, SciPy